

Task-2: Phishing Email Detection & Awareness System

CIN ID: FIT/APR26/CS7871

Project Description:

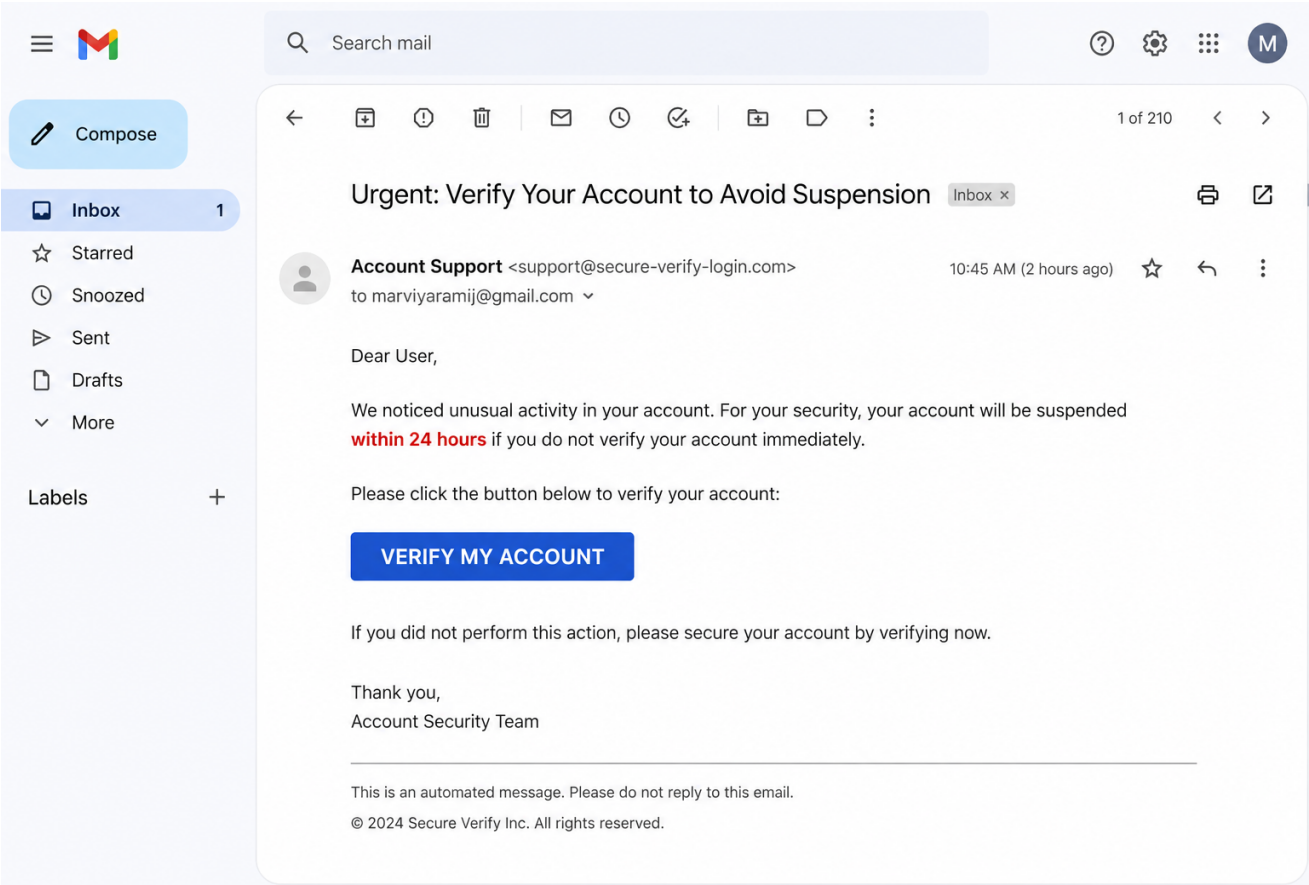
This project demonstrates phishing email detection and cybersecurity awareness techniques. The report analyzes phishing indicators such as spoofed sender domains, urgent language, suspicious links, and fake verification requests. It also explains how phishing attacks attempt to steal sensitive user information using social engineering methods.

The project includes phishing email examples, VirusTotal email analysis screenshots, and awareness guidelines to help users identify and prevent phishing attacks.

Project Objectives

✓ Identify phishing indicators in suspicious emails
✓ Analyze phishing emails using security tools
✓ Classify emails as Safe / Suspicious / Phishing
✓ Provide cybersecurity awareness guidelines

1. Fake Phishing Email Example



The above email demonstrates a phishing attack example using urgent language and a suspicious sender domain. The email attempts to trick users into clicking a fake verification button and revealing sensitive information.

2. VirusTotal Phishing Analysis

Sign in

Sign up

Dashboard

Files

URLs

Search

Hunting

Intelligence

Community

Tools

Premium Services

Sign in

Sign up

14 / 94

14 security vendors flagged this email as malicious

71e8f5b1c1e2f2b7d9c8a6e4f5a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4

email.eml

Size: 25.62 KB

Date: May 22, 2025 12:44 PM UTC

EML

email phishing malicious social-engineering

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 1

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Do you want to automate checks?

Avira	Phishing/Banker.EH	Baidu	Phishing
BitDefender	Phishing.Generic.350962	CrowdStrike Falcon	Phishing (SCAM)
ESET	Phishing.Stealer	Forcepoint ThreatSeeker	Phishing
Fortinet	Phishing.Stealer.Email	Google	Phishing
Kaspersky	HEUR:Phish.Email.Stealer.gen	Lionic	Phishing
Microsoft	Phishing.Email	Netcraft	Malicious Phishing
Sophos	Phish/Email	TrendMicro	Phishing.Email.Stealer
VIPRE	Phishing.Email	Others	No detection

Email Details

From	Account Support <support@secure-verify-login.com>	Date	May 22, 2025 10:45 AM (GMT +05:30)
To	marviyaramij@gmail.com	Message ID	<20250522104523.987654321@secure-verify-login.com>
Subject	Urgent: Verify Your Account to Avoid Suspension	Received	from mail.secure-verify-login.com (185.199.110.153)

VirusTotal analysis shows multiple security vendors detecting the email as malicious or phishing. The report confirms suspicious behavior, malicious indicators, and phishing-related detection results.

3. Awareness & Prevention Guidelines

1. FAKE PHISHING EMAIL (EXAMPLE)

Compose

Inbox

Starred

Snoozed

Sent

Drafts

More

Labels

Urgent: Verify Your Account to Avoid Suspension

Account Support <support@secure-verify-login.com> to marviyaram@gmail.com

Dear User,

We noticed unusual activity in your account. For your security, your account will be suspended **within 24 hours** if you do not verify your account immediately.

Please click the button below to verify your account:

VERIFY MY ACCOUNT

If you did not perform this action, please secure your account by verifying now.

Thank you,

Account Security Team

This is an automated message. Please do not reply to this email.

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2. PHISHING EMAIL ANALYSIS (VIRUSTOTAL)

14 security vendors flagged this email as malicious

71e8f5b1c1e2f2b7d9c8a6e4f5a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2a3f4

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DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY

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Security vendors' analysis

Avira	Phishing/Banker.EH	Baidu	Phishing
BitDefender	Phishing.Generic.350962	CrowdStrike Falcon	Phishing (SCAM)
ESET	Phishing.Stealer	Forcepoint ThreatSeeker	Phishing
Fortinet	Phishing.Stealer.Email	Google	Phishing
Kaspersky	HEUR-Phish.Email.Stealer.gen	Lionic	Phishing
Microsoft	Phishing.Email	Netcraft	Malicious Phishing
Sophos	PhishyEmail	TrendMicro	Phishing.Email.Stealer
VIPRE	Phishing.Email	Others	No detection

Email Details

From	Account Support <support@secure-verify-login.com>	Date	May 22, 2025 10:45 AM (GMT +05:30)
To	marviyaram@gmail.com	Message ID	<20250522104523.987654321@secure-verify-login.com>
Subject	Urgent: Verify Your Account to Avoid Suspension	Received	from mail.secure-verify-login.com (185.199.110.153)

3. AWARENESS & PREVENTION GUIDELINES

How to Identify & Avoid Phishing Emails

1. Check Sender Email Address

Always verify the sender's email address. Phishers use fake or look-alike domains.

support@secure-bank.com

support@secure-bank.com

2. Look for Urgency or Threats

Phishing emails create fear or urgency to trick you into taking action quickly.

3. Check Links Before Clicking

Hover over links to see the actual URL. Don't click on suspicious or shortened links.

http://

4. Avoid Sharing Sensitive Information

Legitimate companies will never ask for your password, PIN, OTP or credit card details via email.

5. Verify Through Official Channels

If an email looks suspicious, contact the organization using the official website or trusted number.

6. Enable Multi-Factor Authentication (MFA)

Enable MFA on your accounts to add an extra layer of security.

Stay Alert, Stay Safe! Think before you click.

Awareness Guidelines for Users

- ✓ Verify sender email addresses carefully
- ✓ Avoid clicking unknown or suspicious links
- ✓ Never share OTPs, passwords, or sensitive information
- ✓ Enable Multi-Factor Authentication (MFA)
- ✓ Use antivirus and email filtering tools
- ✓ Report suspicious emails to security teams

Conclusion

This project successfully demonstrates phishing email detection and cybersecurity awareness techniques. By identifying suspicious email behavior and understanding phishing indicators, users can reduce the risk of credential theft, malware infection, and financial fraud.

Prepared for Cyber Security Awareness & Phishing Detection Task Submission